Bachelor's Thesis Proposal: **Load-Aware Adaptive Thresholding for Change Detection in Continuous IoT Data Streams**

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Continuous data stream processing has become a central challenge in modern data-intensive systems, driven by the rapid growth of real-time applications and Internet of Things deployments. In such settings, large volumes of sensor data are generated. Prior work has shown that these sensors often produce only minor variations between consecutive measurements. As a result, a substantial portion of the data carries limited new information, yet is still processed and stored, leading to high resource utilization.

To address this issue, techniques for reducing data volume have been developed, in particular through change detection mechanisms that forward only data points that differ significantly from previous observations. A prominent approach is the CUSUM [1] (Cumulative Sum) algorithm, which aggregates small deviations over time and triggers a signal once a predefined threshold is exceeded. Furthermore, Mukherjee et al. [2] introduced a mechanism for the real-time adaptation of decision thresholds. By using symbolic dynamic filtering, their approach enables an estimation of thresholds based on current time-series data, thereby allowing for a dynamic threshold in comparison to the static nature of CUSUM.

However, these techniques do not take the current load of the system in which they operate, such as processing load, into consideration. As a result, existing approaches may not effectively utilize available resources to maximize the accuracy of the processed data under varying runtime conditions. In particular, they may discard too much information under low system load, leading to an unnecessary loss in accuracy, or process excessive redundant data under high load, resulting in inefficient resource usage. This raises the question of how high the benefit of incorporating system-level information, such as CPU utilization, into the change detection process is and which dimensions benefit in particular.

Therefore, this thesis aims to investigate whether dynamically adapting change detection thresholds based on the current system state enables a more efficient use of available resources in order to maximize the accuracy of the processed data. In this context, accuracy is understood in terms of the deviation from the original data stream, for example measured via reconstruction error or the ability to preserve significant changes in the signal. The central objective is to analyze whether incorporating system-level information allows for a more effective allocation of computational resources across varying workload conditions.

To answer this research question, a load-aware change detection mechanism will be designed that dynamically adapts its threshold based on system-level metrics, such as CPU utilization, and online stream statistics to calculate the running mean and standard deviation. The study will explore the effects of decreasing the threshold for insignificant changes linearly and exponentially with lower CPU utilization. The approach will be implemented within a simulated streaming pipeline in which system resources, such as CPU and memory, are kept constant while the incoming workload varies. This setup enables a controlled analysis of how effectively the proposed method utilizes available resources under different levels of system load. The evaluation will compare the load-aware approach against established baselines, including static threshold-based and data-driven adaptive methods. Performance will be assessed using metrics such as reconstruction error, detection accuracy of significant changes in the data stream, throughput, and latency, allowing for a systematic evaluation of whether the proposed approach improves accuracy while efficiently utilizing available system resources.

References

[1] Subhadeep Bardhan, Satyajit Swain, Mamata Jenamani, and Aurobinda Routray. Novel cusum methods for repetitive change detection in sensor signals. In *2022 IEEE 1st Industrial Electronics Society Annual On-Line Conference (ONCON)*, pages 1–7, 2022.

[2] Kushal Mukherjee, Asok Ray, Thomas Wettergren, Shalabh Gupta, and Shashi Phoha. Real-time adaptation of decision thresholds in sensor networks for detection of moving targets. *Automatica*, 47(1):185–191, 2011.